

EDUCATION:

Brown University, Providence, RI | *Master of Science, Computer Science*

(Fall 2024 - May 2025)

Brown University, Providence, RI | *Bachelor of Science, Computer Science*

(Fall 2020 - May 2024)

WORK EXPERIENCE:

- **C++ Software Engineer, Dassault Systemes – Johnston, RI** (June 2025 - Present)
 - Ship critical fixes and core optimizations to enterprise clients, reducing application latency and improving reliability.
 - Packaged hypervisor codebase into portable modules to facilitate efficient resource management.
 - Lead development of JavaScript interaction tests simulating end users, increasing test coverage by 50%.
 - Develop Python trace analysis tool to identify performance gaps, automating a previously time-intensive task for the team.
 - Lead design of internal LLM to assist with error triage and expedite risk assessment.
- **Fullstack Software Engineer Intern, LI-COR (Formerly Onset Data Loggers) – Bourne, MA** (May 2024 - August 2024)
 - Engineered MQTT handler processing 10k+ IoT messages/day, accelerating device-to-cloud communication.
 - Built full-stack cloud platform to consolidate and manage firmware files, significantly reducing manual configuration time.
 - Prototyped a weather map dashboard with Leaflet and GIS APIs, enabling real-time user interaction with live weather data.
- **Head Teaching Assistant, Brown University CS – Providence, RI** (October 2023 - May 2024)
 - Led 14-person TA team for 100+ students, managing grading, office hours, and course materials for an upper-level CS course.
 - Redesigned 6 core projects to more accurately reflect modern industry UI/UX principles.
 - Coordinated portfolio reviews with 11 industry professionals, to provide practical feedback for students.
- **Frontend Software Engineer Intern, LI-COR (Formerly Onset Data Loggers) – Bourne, MA** (May 2023 - August 2023)
 - Created a notification interface for users to monitor various weather sensors in line with the company's new design library.
 - Overhauled product sorting functionality as part of company-wide user interface redesign.
 - Improved middleware API efficiency as a part of the core engineering team, reducing overall latency for end users.
- **Founder & Content Creator, Independent Art Business – tiktok.com/@minghuuaa** (January 2020 - December 2023)
 - Grew a digital art brand to 170k+ followers, gaining millions of views and engagement through A/B testing and other analytics.
 - Sold 300+ digital prints and custom merchandise generating \$1.6k profit in 3 months.
 - Secured official brand partnerships with several digital drawing tablet companies such as Huion and XPPen.

RECENT PROJECTS:

- **Ninja Mouse (Custom 3D Game Engine)**
 - Developed a C++ game engine with first and third person control, collision detection, custom shaders, and AI pathfinding.
 - Designed a component-based system to allow game-specific customizations.
 - Integrated custom environment shaders and 3D assets with material loading support.
 - Leveraged BVH and grid-based acceleration trees to optimize collision detection between static and dynamic objects.
- **Origin Two (Custom 2D Game Engine)**
 - Built a Java game engine with a reusable UI and sound toolkit, animation cycling, and physics and collision systems.
 - Scaled engine to support 3 dungeon crawler and platformer games with additional procedural level loading and NPC AI.
- **Realtime Ocean Water Simulation**
 - Implemented a physically accurate ocean water simulation based on motion-picture oceans.
 - Built a fast fourier transform optimizer to allow real-time user manipulation of wind speed and turbulence.
 - Introduced new features ocean foam, particle systems for spray, caustics, and custom shaders for reflection and refraction.
- **Realtime Graphics & Ray Tracing**
 - Created a real-time C++/OpenGL scene viewer with shape and camera manipulation and custom GLSL shaders.
 - Integrated ray tracing algorithm for realistic rendering using the Phong model, mirror reflections, and texture mapping.
 - Programmed custom toon and outline shaders with framebuffer objects to achieve stylized post-processing effects.
- **3D Modeling & Animation**
 - Animated game-ready and cyclable attack sequences with a custom 3D rig and character in Unreal engine.
 - Designed, modeled, rigged, textured, and animated 2 original characters with custom shaders for an animated short.
 - Created adaptable shader and texture customization sliders, supporting rapid creation of varying environments.

SKILLS:

- **Computer Languages:** C++, Python, Java, OpenGL, GLSL, TensorFlow, React.js, Vue.js, Node.js, JS, TS, Shell, HTML/CSS
- **Tools:** Unreal Engine 5, Maya, Blender, Gaea, Adobe, Git, Docker, Figma
- **Skills:** Atlassian/Jira, Agile, CI/CD, Scrum, Kanban, Project Management